AI RESEARCHER

Inha Kang

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Scholar · GitHub · LinkedIn
Portfolio

Vision-Language Models | VLM Foundation Models | Explainable AI | Physical Understanding | World Models | Agentic Tool Calling

I turn insights from model failures into more capable and reliable AI. My research on vision-language and foundation models connects interpretability with better data, representations and learning objectives across perception, physical prediction and agentic tool use.

Research & Experience

KRAFTON Research Intern — 31 Aug 2026 – Present
LLM foundation models and agentic tool calling — Korea

- Evaluated **BFCL v4** and **tau3-bench**; diagnosed model failures separately from parser, serving and evaluation-protocol effects.
- Audited mid-training/SFT data; developing **role-aware tool-use curation** with schema checks and consistent training/serving formats.
- Designing **executable RL tasks**, reference replay and state-based verifiers for post-training.

KAIST Ph.D., Artificial Intelligence — Aug 2024 – Present
Vision-language models, foundation models and explainable AI — Seoul, Korea

- Probe **representation and reasoning failures** to explain spatial, compositional and language biases in vision-language models.
- Develop methods for **visual grounding, robust perception and foundation-model adaptation** through representation learning and decision-aware objectives.
- Publications at **CVPR, ICLR and ECCV (2026)**; ongoing **world-model** research.

FaindersAI Deep Learning Researcher — 2022 – 2024
Data-centric AI and multimodal perception — Seoul, Korea

- Built **VLM-based segmentation-data curation** and real-time retail change detection.
- Researched continual learning and **tabular fusion for heterogeneous sensors**.

Education

KAIST Ph.D., Artificial Intelligence — 2024 – 2028 (expected)
CVML Lab · Advisor: Prof. Hyunjung Shim · GPA: **4.3/4.3**

KAIST M.S., School of Computing — 2020 – 2022
CGV Lab · Advisor: Prof. Jinah Park · GPA: 3.88/4.3

GIST B.S., Electronic Engineering & Computer Science — 2016 – 2020
Salutatorian · GPA: 3.94/4.5 · Minor: Business Administration and Sociology

UC Berkeley Visiting study, Computer Science — 2017

Selected Publications

Vision-language understanding, foundation models and physical prediction. * Equal contribution.

ECCV
2026
Paper ↗

Blind to Position, Biased in Language: Probing Mid-Layer Representational Bias in Vision-Language Encoders for Zero-Shot Language-Grounded Spatial Understanding

N. M. An*, **I. Kang***, M. Lee, H. Shim

Vision-Language Models · Explainable AI · Spatial Reasoning

Probed positional and language biases, then reused mid-layer representations for **training-free grounding and retrieval**.

ICLR
2026
Paper ↗

What "Not" to Detect: Negation-Aware VLMs via Structured Reasoning and Token Merging

I. Kang, Y. Lim, S. Lee, J. Choi, J. Choe, H. Shim

Multimodal Reasoning · Efficient Adaptation · Data-Centric AI

Addressed affirmative bias with structured data, token merging and targeted LoRA; up to **+10.8 NMS-AP** on OVDEval negation.

CVPR
2026
Paper ↗

Real-Time Long Horizon Air Quality Forecasting via Group-Relative Policy Optimization

I. Kang, E. Kim, W. Ryu, J. Shin, S. Yu, Y.-H. Kang, S. Jeong, E. Kim, S. Kim, H. Shim

Foundation Models · Reinforcement Learning · AI for Science

Aligned forecasting with asymmetric decision costs via GRPO; **47.3% lower false-alarm rates** than supervised fine-tuning alone.

Findings of EMNLP
2025
Paper ↗

3D-Aware Vision-Language Models Fine-Tuning with Geometric Distillation

S. Lee*, J. Choi*, **I. Kang**, J. Kim, J. Park, H. Shim

Multimodal AI · 3D Reasoning · Knowledge Distillation

Distilled geometric cues from **3D foundation models** into VLMs to improve spatial reasoning without changing the architecture.

CVPR
2025
Paper ↗

No Thing, Nothing: Highlighting Safety-Critical Classes for Robust LiDAR Semantic Segmentation in Adverse Weather

J. Park, H. Lee, **I. Kang**, H. Shim

3D Perception · Robustness · Domain Generalization

Addressed semantic and local feature corruption to preserve **safety-critical objects** under adverse-weather domain shift.

Current Manuscript

Manuscript in preparation
2026

Physically Conditioned Action and Ordered Uncertainty for Multi-Day PM Forecasting over East Asia

I. Kang*, W. Ryu*, S.-C. Hong, J.-B. Lee, S. Ban, S. Jeong, Y.-H. Kang, S. Kim, H. Shim

Physical Understanding · World Models · Uncertainty

PLUME learns and transfers bounded correction and uncertainty states without recursive field feedback; highest ETS in all six pollutant/region settings in 2024.

Additional Publications

Earlier research in model evaluation, 3D imaging and anomaly detection.

CVPR
2023
Paper ↗

Why is the winner the best?

M. Eisenmann, A. Reinke, V. Weru, . . . , **I. Kang**, et al.

Model Evaluation · Failure Analysis · Data-Centric AI

Contributed to a multi-center study of **80 biomedical-imaging competitions**, examining successful development strategies and the role of failure-case analysis.

Journal of Hazardous Materials
2023
Paper ↗

Three-dimensional label-free visualization of the interactions of PM2.5 with macrophages and epithelial cells using optical diffraction tomography

W. S. Lee*, **I. Kang***, S. J. Yoon, et al.

3D Imaging · Quantitative Imaging · Biomedical Imaging

Used label-free 3D imaging to quantify cellular $PM_{2.5}$ uptake, refractive index and mass changes.

MICCAI MOOD Challenge
2022
Paper ↗

Joint embedding of 2D and 3D networks for medical image anomaly detection

I. Kang, J. Park

Anomaly Detection · Representation Learning · 3D Vision

Joint 2D/3D feature learning for brain MRI and abdominal CT; **1st place in the sample-level task**.

KTCP
2022
Paper ↗

End-to-End Vertebra CT Image Segmentation Network with the 3D Surface-Enhanced Module and the Trainable Preprocessing Method

I. Kang, J.-H. Cho, J. Park

3D Vision · Segmentation · Medical AI

Integrated trainable preprocessing and surface-enhanced features for end-to-end vertebra segmentation.

MICCAI MOOD Challenge
2021
Paper ↗

Self-supervised 3D out-of-distribution detection via pseudoanomaly generation

J.-H. Cho*, **I. Kang***, J. Park

Self-Supervised Learning · Synthetic Data · OOD Detection

Learned anomaly detection from normal data using synthetic pseudoanomalies; **1st place**.

Honors & Awards

CVPR Compute Transparency Champion Award 2026

MICCAI MOOD Challenge – 1st Place 2021, 2022
Two consecutive years

Best Presentation Award 2022
KTCP

Best Thesis Award 2020
GIST

Earlier Research & Engineering

Machine Learning and Vision Lab Research Intern 2019 – 2020
Gwangju, Korea
Studied scheduled training for long-tailed object detection in autonomous driving.

Aerospace Propulsion Lab Research Intern 2017 – 2018
Gwangju, Korea
Worked on parallel computing and GPU acceleration.

SketchScope: Interactive Volume Visualization ↗
Mar 2021 – Aug 2022

Developed sketch-based volume exploration with hand-gesture and touch interaction across Leap Motion, VR and holographic displays. Extended the platform to HoloLens 2 with QR-code-based volume placement.

ForMixMatch: Virtual Fashion Coordinator ↗
Sep – Dec 2020

Built a VITON-based service for combining upper- and lower-body apparel. Compared four interface layouts in a user study to assess mix-and-match interaction and user confidence.

AI Health Trainer ↗
Apr – Dec 2018

Built real-time mobile posture correction using OpenPose and Realtime PoseNet. Led the GIST representative team for JETS and presented the project at NAVER.

GPU Acceleration with Kokkos ↗
Dec 2017 – Dec 2018

Implemented parallel workloads in C++ with Kokkos; explored architecture-aware memory access, portable execution patterns and GPU work mapping.

Technical Expertise

Multimodal learning & model analysis
Vision-language grounding; layer-wise representation probing; attention analysis; parameter-efficient adaptation (LoRA); 3D perception; self-supervised learning.

LLM & agentic systems
Mid-training data audits; SFT data curation; multi-turn tool-use trajectories; schema validation; deterministic replay; benchmark and failure analysis.

Physical prediction & learning
Task-specific world models; physics-guided learning; uncertainty estimation; decision-aware reward design; reinforcement learning (GRPO).

Research engineering
Python; C++; GPU computing; Kokkos; data validation and provenance; reproducible evaluation pipelines.

Invited Talk & Patent

Invited Speaker, Korean Society of Medical Imaging and Information (KOSMII), 2022.

Data Collection Apparatus and Method. Korean Intellectual Property Office; Publication No. 10-2025-0031529.